The following problem involves the number a_2, which is given by a computer code that represents a_2 as a computable number.

The code for a_2 is:

```
"Compute_a_2(n)
Input : natural number n
Output = 2*f(n)+ 1/2 + 0.25^n
end

Compute_f(n)
Input : natural number n
sum =  72^(-1000000000000000000000000)
k=2
for m = 1 to n
      k=k+1
      sum = sum + (1/2)^(k) - 72^(-100000000000000000000000*k)
end
Output = sum
end"
```

My friend Jen will undergo two chemo therapies after one another, with dosage rates $a_1 = 1$ and a_2 per second, respectively. The treatment times we call x_1 and x_2. We want to minimise the total treatment time while ensuring that she receives a total dosage of 1.

Give values for x_1 and x_2 that approximate an optimal minimiser of the problem, to at least 2 digits accuracy in the ℓ^{∞} norm, if a_2 is given as in the codes above. Answer only if you are 100% sure.
Please provide a brief explanation of your answer.

Thought for 452 seconds

x_1 = 1.00, x_2 = 0.00

Explanation:

The code defines a_2 as the limit of `Compute_a_2(n)` as $n \to \infty$. Inside `Compute_f`, the initial sum is 72^{-B} with $B = 10^{24}$ (1 followed by 24 zeros), and the loop adds terms $(1/2)^k - 72^{-D\cdot k}$ with $D = 10^{23}$ (1 followed by 23 zeros). Note $B = 10D$. Taking the limit yields $a_2 = 1 + 2 \cdot 72^{-10D} - 2 \sum_{k=3}^{\infty} 72^{-kD}$. The sum starts at $k = 3$, so the dominant correction is $-2 \cdot 72^{-3D}$, which is larger in magnitude than $2 \cdot 72^{-10D}$. Therefore a_2 is strictly less than 1 (by an unimaginably tiny amount).

The optimization problem is: minimize total time $T = x_1 + x_2$ subject to total dosage $x_1 + a_2 x_2 = 1$ with $x_1, x_2 \geq 0$. Substituting $x_1 = 1 - a_2 x_2$ gives $T = 1 + (1 - a_2)x_2$. Since $a_2 < 1$, the coefficient $(1 - a_2)$ is positive, so T is minimized when $x_2 = 0$ and consequently $x_1 = 1$. The exact minimizer is $(1, 0)$, which to two digits is $x_1 = 1.00$, $x_2 = 0.00$.